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Introduction

This guide covers:

- Creating a Blank Subject
- Attaching Labelling Skeletons to Subjects
- Creating a Subject from a Labelling Skeleton
- Saving and Loading Subject VSKs

This is the third of the three capture preparation guides:

1. Creating a Database
2. Creating a Labelling Skeleton
3. ***Creating a Subject***

Subject Setup

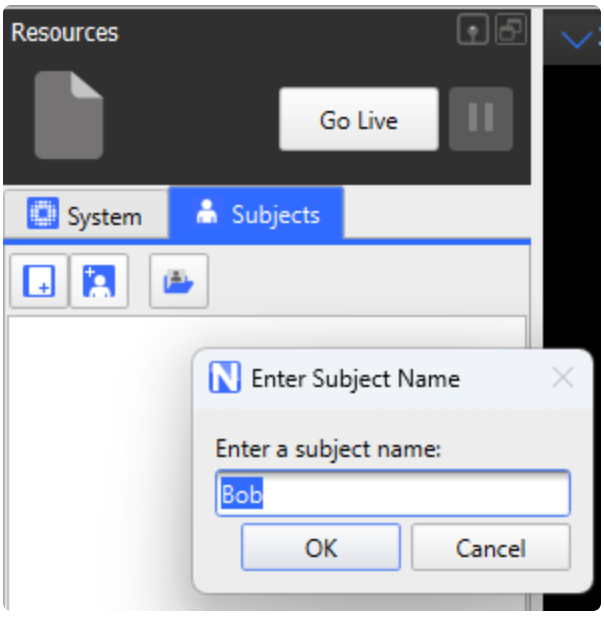
In order to capture 3D data of an individual or object the Vicon Nexus system must recognise it as a subject. This guide will take you through the steps of creating a subject, as either a blank subject to create a labelling skeleton from, or from an existing labelling skeleton.

Terminology:

- **Labelling Skeletons** are attached to **Subjects**.
- **Labelling Skeletons** represent **marker sets**, labelled within Vicon Nexus, as templates for motion capture.
- These templates must then be calibrated to individual subjects (the individual or object that fits the layout).
- A full-body marker suit would be a **labelling skeleton** within Nexus, whereas the individual wearing it would be a **subject** within Nexus; each individual that wears the suit would have different specific marker layouts based on the movement of their body, and therefore need to be individually calibrated to the skeleton.
- Templates save as **VST** files (*Vicon Skeleton Template* file).
- Subjects save as **VSK** files (*Vicon Skeleton* file).

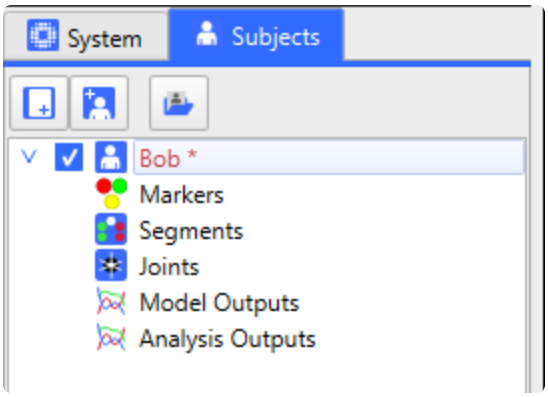
Creating a Blank Subject

Navigate to the **Resources panel** and **subject tab**, then click the '**Create a blank subject**' button and name your subject.



Entering a subject name

This will add a subject without a labelling skeleton.

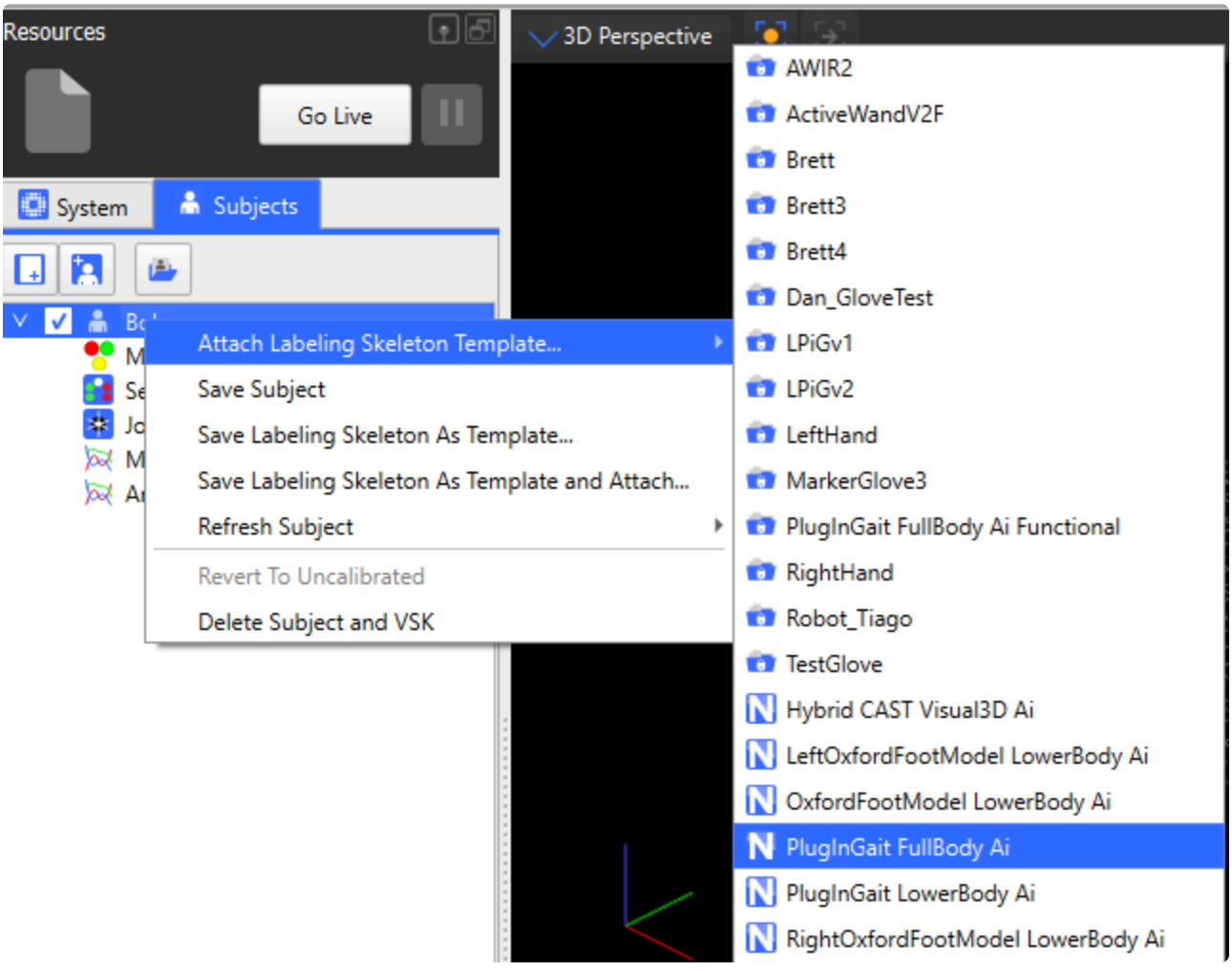


Subject without labelling skeleton

Attaching a Skeleton to a Blank Subject

Subjects need labelling skeletons in order to be tracked.

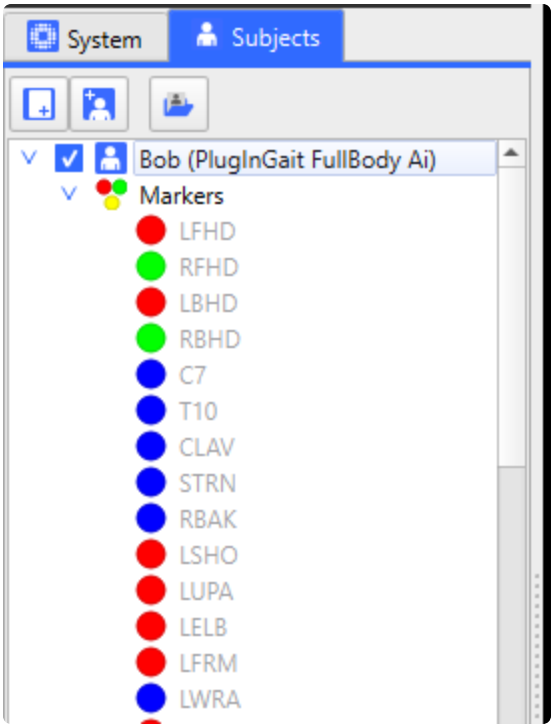
Right click the subject and '**Attach Labelling Skeleton Template**' then select an existing template.



Adding a labelling skeleton template (VST) to a blank subject

For this demonstration we will be using the '**PlugInGait FullBody Ai**' skeleton; attach this to your subject.

You will see that the subject sections (markers, segments, and joints) have now been populated.



Plug-In Gait markers

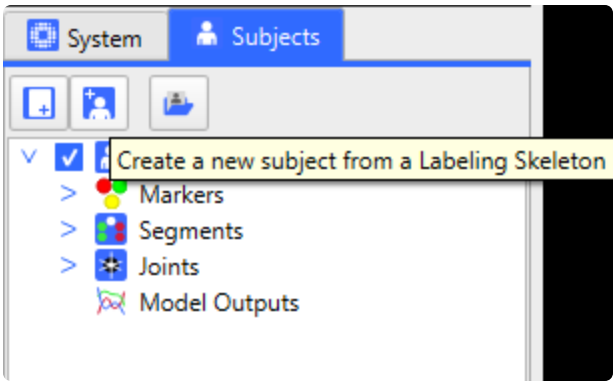
This is now a subject with an existing labelling skeleton.

Once you have applied the skeleton you will need to calibrate the template against this subject.

Creating a Subject from a Labelling Skeleton

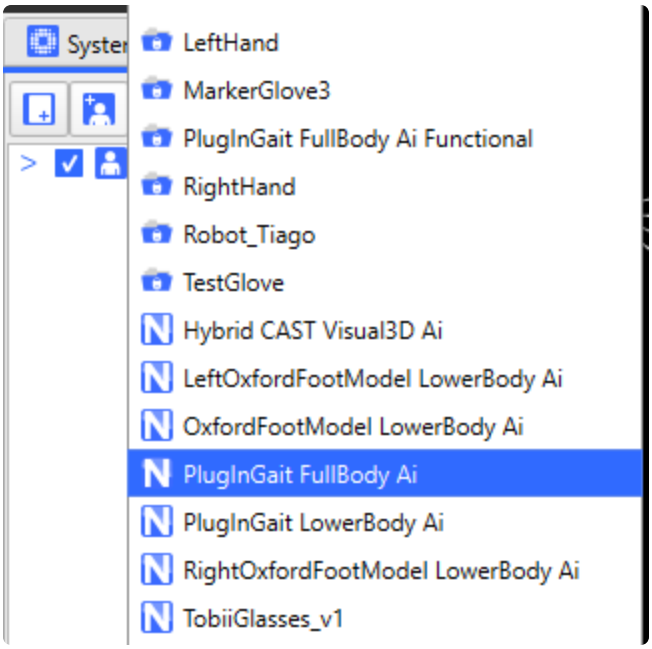
To create a new subject from a labelling skeleton:

- Navigate to the **Resources panel** and the **subject tab** within.
- Click the '**Create a new subject from a Labelling Skeleton**' button



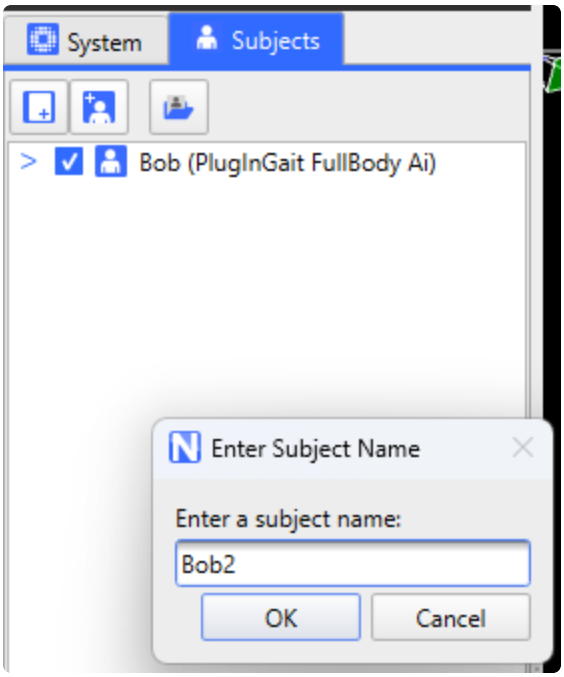
Button create a new subject from an existing labelling skeleton template (VST)

Select the labelling skeleton template for the subject to use.



Selecting which skeleton template to use

Name the subject.



Naming your subject

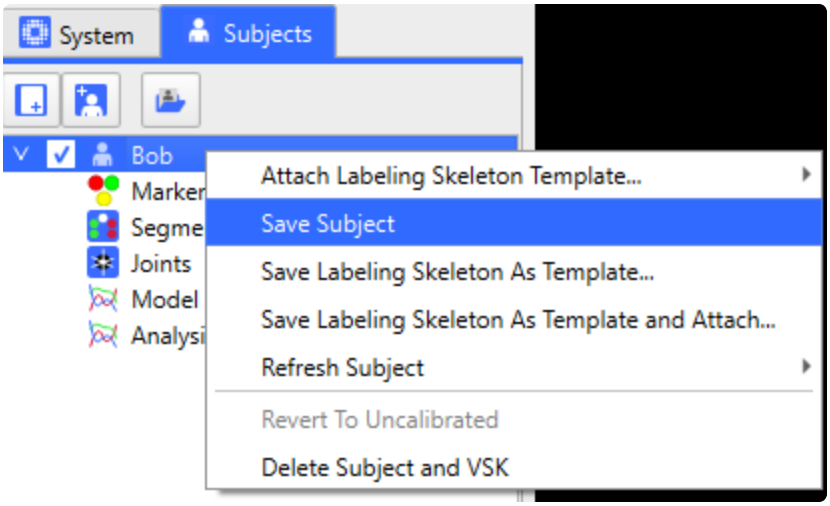
This is now a subject with a labelling skeleton.

Once you have applied the skeleton you will need to calibrate the template against this subject.

Saving and Loading Subject FIles

Save an existing subject

To save an existing subject - right click the subject and click 'Save Subject'.

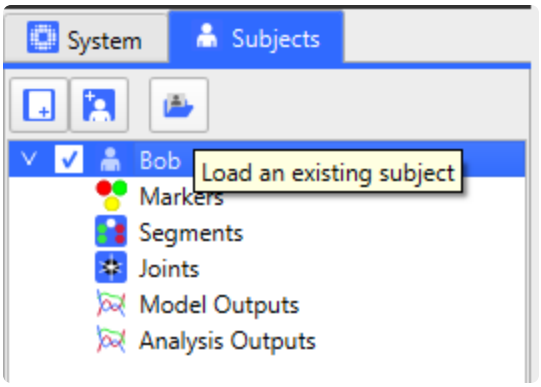


Right click menu options for saving a subject

This will create a VSK (a calibrated VST) within the capture session directory.

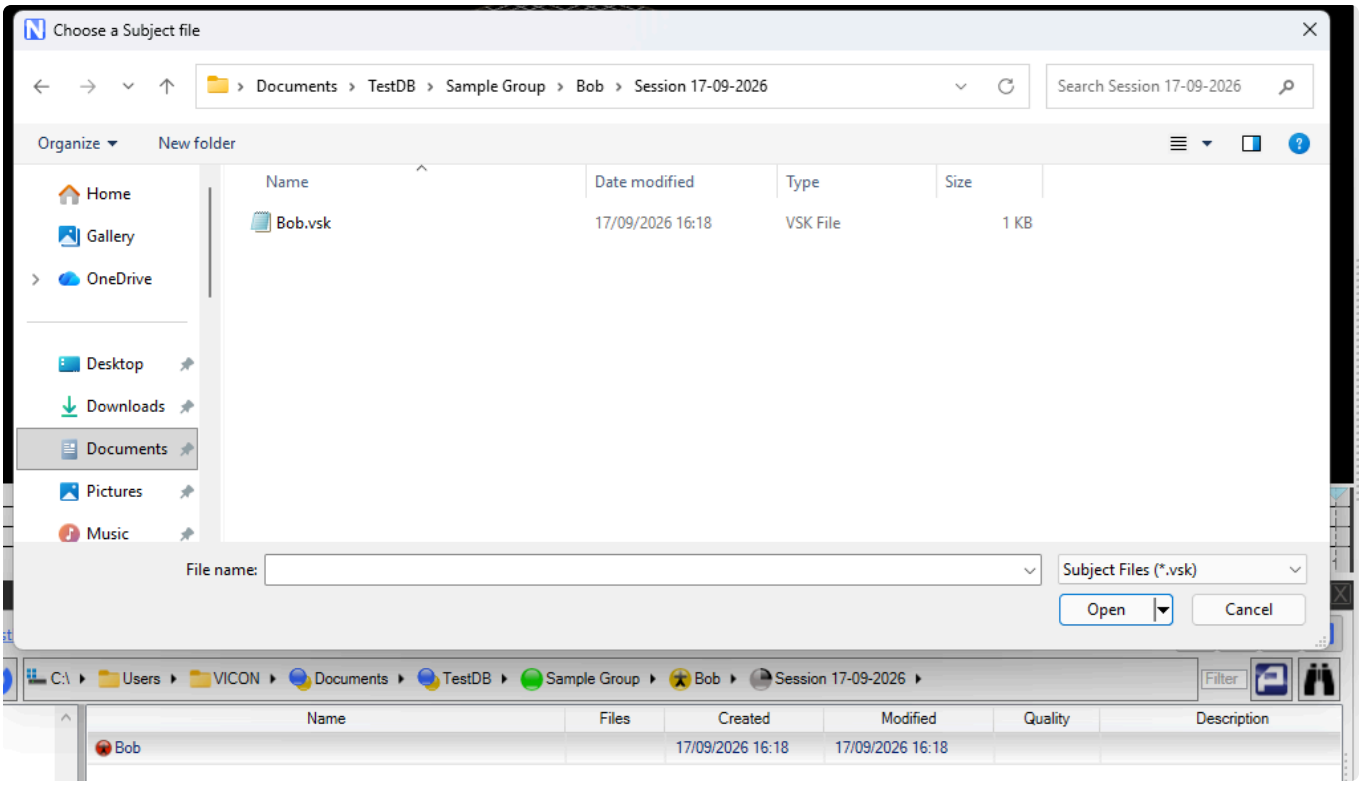
Load an existing subject

From the **Resources panel/Subject tab** you can load an existing subject from a previous save file.



Button to load an existing subject file (VSK)

Navigate to where the capture session directory, within the ProEclipse database, is saved on disk.



Saving the VSK file

This will load the subject, alongside their attached labelling skeleton, to the active setup.